

Daniel Fernandez

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EDUCATION

Queens College, City University of New York (CUNY)

Expected Dec 2027

B.S. Computer Science — GPA: 3.47

Flushing, NY

- Relevant Coursework: Cloud Computing, Operating Systems (C++/C), Databases (SQL), Data Structures & Algorithms (Java), Advanced Python Programming, Software Engineering, Object-Oriented Programming (Java)

EXPERIENCE

Dow Jones & Company

Jun 2026 – Aug 2026

Software Engineer Intern – Wall Street Journal Web Brand

New York, NY

- Lead a cross-functional intern team building an AI-powered testing harness, packaged as a Claude Skill, that maps cross-repo code dependencies and auto-generates self-healing Playwright e2e tests informed by Jira context.
- Merged production code to dynamically swap news buckets on the WSJ homepage, improving content delivery for millions of users.
- Patched a structural rendering issue within a shared component library, resolving a bug that injected duplicate HTML tags and ensuring correct semantic and SEO baseline compliance across multiple consuming products.

CUNY Tech Prep

Jul 2025 – Jun 2026

Software Engineer Fellow - Web Dev Track

New York, NY

- Developed an educational web app with React, Next.js, and Node.js tailored to boost college student productivity.
- Leveraged 15+ REST API endpoints to fetch and manage data for user-facing features.
- Worked in agile team workflows using Git/GitHub, code reviews, Scrum routines, and CI/CD practices.

Google X BASTA

Mar 2026 – Jun 2026

Software Engineer Mentee

Remote

- Selected for the Google Software Engineering Mentorship Program, receiving weekly 1:1 mentorship from a Google engineer to strengthen data structures, algorithms, and technical interview problem-solving skills.

PROJECTS

codeburrow | *Python, tree-sitter, Git, VoyageAI, ChromaDB*

Jul 2026 – Aug 2026 | [GitHub](#)

- Built and packaged a pip-installable hybrid code search CLI, fusing BM25 keyword search with Voyage AI semantic embeddings via Reciprocal Rank Fusion to improve retrieval relevance over keyword or vector search alone.
- Designed a multi-language AST chunking pipeline with tree-sitter (Python, JavaScript, TypeScript/TSX), parsing source files into function- and class-level chunks for higher-precision embedding and retrieval.
- Implemented git-aware incremental indexing via a self-installing post-commit hook, re-embedding only changed files and safely extending a repo's existing hook interpreter (shell, Python, Node, Ruby).

codejam | *Java, Spring Boot, Docker, React, TypeScript, WebSockets*

Aug 2026 – Present | [GitHub](#)

- Built a real-time collaborative code editor (React, TypeScript, Monaco, Yjs) supporting simultaneous multi-user editing without conflicts or lost changes.
- Architected a Java backend that orchestrates Docker containers for sandboxed code execution, using multithreading, concurrent process management, and timeout handling to safely run untrusted code via a Spring Boot REST API.
- Designed a Spring Boot WebSocket service to relay real-time collaborative edits across all connected users, using a lightweight architecture that kept the system self-contained without adding extra backend infrastructure.

M.A.P.S. | *React, Node.js, PostgreSQL, Firebase*

Sep 2025 – Dec 2025 | [GitHub](#)

- Architected a full-stack academic planning platform designed to scale for roughly 2,000 Queens College undergraduates, centralizing course mapping, professor metrics, and schedule creation.
- Designed relational data models and built REST API routes using Node.js/Express, PostgreSQL (Neon), and Drizzle ORM.
- Integrated real-time course data from CUNY Global Search and implemented user-scoped operations with Firebase Authentication.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, SQL (Postgres), C++, C, HTML/CSS

Frameworks: Next.js, React.js, Spring Boot, Node.js, GraphQL, Playwright

Developer Tools: Git, Claude Code, Docker, VS Code